

Inclusive Monetary Policy: How Tight Labor Markets Facilitate Broad-Based Employment Growth

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Motivation

*With regard to the employment side of our mandate, our revised statement emphasizes that maximum employment is a **broad-based** and **inclusive** goal. This change reflects our appreciation for the benefits of a strong labor market, particularly for many in low- and moderate-income communities.*

— Jerome Powell, 2020 Jackson Hole Economic Policy Symposium

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- What's monetary policy's heterogeneous effects on different segments of the labor markets?

This paper

- Study how labor market strength **intermediates** the effect of monetary policy across different types of workers and demographic groups.
 - Exploit **regional variation** in employment-population ratios.
 - Large and persistent effects on the employment growth rate of **less-attached** workers (i.e. Blacks, high school non-completers, women) in tight labor markets.

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- Study how labor market strength **intermediates** the effect of monetary policy across different types of workers and demographic groups.
 - Exploit **regional variation** in employment-population ratios.
 - Large and persistent effects on the employment growth rate of **less-attached** workers (i.e. Blacks, high school non-completers, women) in tight labor markets.
- Develop a NK model with **heterogeneous workers** to rationalize the empirical results.
 - Two *ex-ante* different groups of workers + ex-post within-group heterogeneity in labor productivity + fixed cost of hiring
 - MP affects hiring/firing thresholds differentially $\Rightarrow$ heterogeneous responses

US Census Bureau's Quarterly Workforce Indicators (QWI) program

- Quarterly local labor market level employment statistics for industry-worker demographics cells.
- Period: Q1 1990 - Q1 2019
- 895 local labor markets = 380 Metropolitan Statistical Areas and 515 Micropolitan Statistical Areas
- 2-digit NAICS industry (e.g. Manufacturing, Wholesale Trade, Professional, Scientific, and Technical Services)
- Drop MSA-industry-group-quarter cells that have less than 50 employees.

Labor force attachment by demographic groups

Table 1: Average Labor Force Attachment by Demographic Group, 1990q1–2019q1

	Mean	Standard Error
Blacks	56.6%	0.1
Whites	62.3%	0.1
Less than High School	40.3%	0.1
High School	58.9%	0.2
Some College	68.1%	0.2
Bachelors Degree	75.7%	0.1
Female	55.2%	0.1
Male	68.5%	0.2

Data are calculated from statistics reported by the Bureau of Labor Statistics.

Summary statistics

Table 2: Summary Statistics

	N	Mean	SD	p10	p25	p50	p75	p90
Federal Funds Rate	1,204,974	2.32	2.25	0.09	0.16	1.52	4.81	5.42
Monetary Shock	1,204,974	-3.73	0.93	-4.58	-4.57	-3.70	-3.59	-2.19
Emp/Pop	1,204,974	0.67	0.14	0.49	0.58	0.67	0.77	0.86
<i>Two Year Employment Growth</i>								
Blacks	513,140	10.04	21.81	-12.71	-2.01	8.01	18.84	33.75
Whites	1,019,587	6.12	13.72	-7.55	-0.98	4.76	11.23	20.67
Less than High School	753,583	2.12	14.09	-12.68	-5.67	0.92	8.35	17.84
High School	1,031,445	0.60	12.54	-12.09	-6.08	-0.59	5.69	14.18
Some College	1,039,754	0.97	12.37	-11.53	-5.55	-0.24	5.88	14.35
Bachelors Degree	920,562	1.12	12.02	-11.39	-5.36	0.08	6.14	14.31
Female	1,082,355	6.53	15.74	-9.48	-1.67	5.04	12.51	23.34
Male	1,155,480	7.02	15.84	-8.76	-1.20	5.46	12.84	23.66

This table provides summary statistics for the main variables used in the analysis. The statistics are equal-weighted across MSA-industry-subgroup-quarter cells.

Main specification

$$\begin{aligned} EmplGrowth_{j,g,m,t} = & \beta_1 \times FedFunds_t \times Empl/Pop_{m,t-1} \\ & + \beta_2 \times Empl/Pop_{m,t-1} + \theta_{j,m} + \delta_{j,t} + \epsilon_{j,g,m,t} \end{aligned} \quad (1)$$

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- $\theta_{j,m}, \delta_{j,t}$: industry-MSA FE and industry-time FE

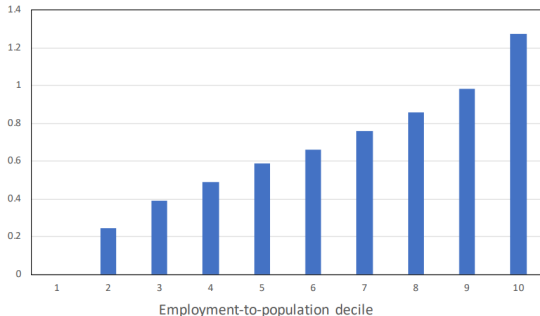
Results

Table 3: Two-Year Employment Growth and Federal Funds Rate by Labor Market Tightness

Panel A: Race				
	(1) Blacks	(2) Whites		
Fed Funds Rate × Emp/Pop	-1.09*** (0.40) [0.00]	0.10 (0.18)		
R ²	0.30	0.28		
Observations	511,843	1,019,176		
Panel B: Education				
	(3) Less than High School	(4) High School	(5) Some College	(6) Bachelors Degree
Fed Funds Rate × Emp/Pop	-0.47** (0.20) [0.00]	0.00 (0.17) [0.66]	0.02 (0.16) [0.77]	0.05 (0.17)
R ²	0.30	0.26	0.26	0.27
Observations	752,685	1,030,813	1,039,149	919,853
Panel C: Sex				
	(7) Female	(8) Male		
Fed Funds Rate × Emp/Pop	-0.26 (0.18) [0.02]	-0.03 (0.20)		
R ²	0.28	0.24		
Observations	1,081,865	1,155,071		

Heterogeneous effects

Figure 1: Predicted Black Employment Growth by Labor Market Tightness, Fourth Quarter 2000



- **Experiment:** 1-SD decrease (2.25pp) in the federal funds rate
- **y-axis:** two-year employment growth rate (in pp) *relative to the first decile*
- **Statistics:** mean two-year black employment growth rate is 10.04%, 90th-percentile of E-P ratio is 86%, 10th-percentile is 49%.

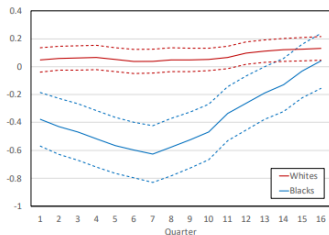
Temporal dynamics of the heterogeneous effects

Goal. Want to understand the short-run and long-run dynamic of the heterogeneous responses. Two exercises:

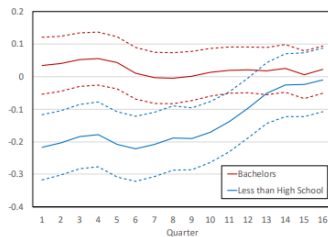
- ① **One-year rolling window:** Re-estimate (1) for the growth rate of employment from the beginning of quarter $t + p$ through the end of quarter $t + p + 3$, where $p \in [1, 16]$
- ② **Cumulative effect:** Re-estimate (1) for the growth rate of employment from the beginning of quarter $t + 1$ through the end of quarter $t + 1 + s$, where $s \in [1, 16]$.

Results: one-year rolling window

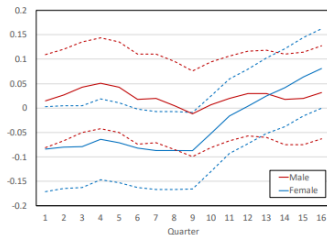
Panel A. Race



Panel B. Education

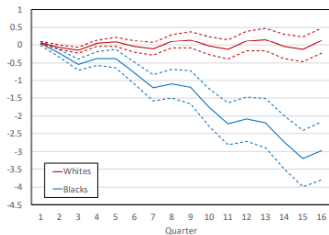


Panel C. Sex

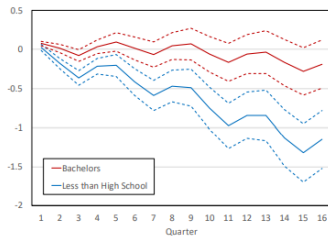


Results: cumulative effect

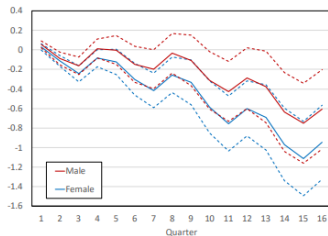
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Panel B. Education



Panel C. Sex



High-frequency monetary shocks

Issue. Decreases in the federal funds rate tend to occur in response to deteriorations in the economy + employment growth in *slack* labor markets is *more* pro-cyclical = upward bias (i.e. less negative)

$$\begin{aligned} EmplGrowth_{j,g,m,t} = & \beta_1 \times FedFunds_t \times Empl/Pop_{m,t-1} \\ & + \gamma \times BusinessCycle_t \times Empl/Pop_{m,t-1} + \dots \end{aligned}$$

$$Cov(FedFunds_t, BusinessCycle_t) > 0$$

$$\gamma < 0$$

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$$Cov(FedFunds_t, BusinessCycle_t) > 0$$

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Solution. Instrument with high-frequency monetary shocks construction

$$\begin{aligned} FedFunds_t \times Empl/Pop_{m,t-1} = & \alpha_1 \times MonetaryShock_t \times Empl/Pop_{m,t-1} \\ & + \alpha_2 \times Empl/Pop_{m,t-1} + \theta_{j,m} + \delta_{j,t} + \eta_{j,g,m,t} \end{aligned}$$

Results: high-frequency monetary shocks (2SLS)

Table 5: **Two-Year Employment Growth and Federal Funds Rate by Labor Market Tightness: 2SLS Estimates**

Panel A: First Stage

	(1) Fed Funds Rate × Emp/Pop
Monetary Shock × Emp/Pop	2.13*** (0.03)
<i>F</i> – statistic	4,984.19
Observations	511,843

Panel B: Race

	(2) Blacks	(3) Whites
Fed Funds Rate × Emp/Pop	-1.23** (0.51) [0.00]	0.05 (0.24)
<i>R</i> ²	0.00	0.01
Observations	511,843	1,019,176

Panel C: Education

	(4) Less than High School	(5) High School	(6) Some College	(7) Bachelors Degree
Fed Funds Rate × Emp/Pop	-0.66*** (0.25) [0.00]	-0.15 (0.22) [0.58]	-0.17 (0.22) [0.42]	-0.08 (0.25)
<i>R</i> ²	0.01	0.01	0.01	0.00
Observations	752,685	1,030,813	1,039,149	919,853

Panel D: Sex

	(8) Female	(9) Male
Fed Funds Rate × Emp/Pop	-0.44* (0.25) [0.07]	-0.22 (0.27)
<i>R</i> ²	0.00	0.01
Observations	1,081,865	1,155,071

Taking stock

- ① Monetary expansions have a larger effect on the employment growth of less-attached workers in tight labor markets.
 - **Sizable differences:** 9% and 18% of mean 2-year employment growth rates for Blacks and high-school non-completers, when comparing the tight (90th percentile) labor market and the slack (10th percentile) labor market
 - **Long lasting:** differences in cumulative employment growth persists even after 4 years.

- ② The labor market tightness channel **only** presents for less-attached workers.

Overview of the model

- Textbook RANK model + heterogeneous worker
 - CES monopolistic competition + Calvo pricing
 - Habit formation
 - No capital
 - Log-linearization
- Two types of workers, whose worker-specific productivity are drawn from **different** subsets of the unit interval
 - Different levels of average productivity map into different levels of steady state employment between the two types.
- Hiring friction + fixed firing cost = endogenous type-specific hiring and firing thresholds

Household

- A representative household consists of a continuum of workers of two types, **high (h)** and **low (l)**
 - Mass γ of high types, mass $1 - \gamma$ of low types
 - A high type's productivity is drawn *uniformly* from the interval $[\underline{s}, 1]$, while a low type draws from $[0, \bar{s}]$.
- Household's utility is given by

$$U_t = \frac{1}{1 - \sigma} (C_t - hC_{t-1})^{1 - \sigma} - (N_{h,t}^\chi + N_{l,t}^\chi) / \chi,$$

where $N_{k,t}$ is the number of type- k workers working in period t .

- Labor FOC:

$$\frac{N_{k,t}^{\chi-1}}{Z_t} = \frac{W_{k,t}}{P_t},$$

where Z_t is the marginal utility of consumption.

Labor market: timeline

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- ④ **Hiring.** Firms employ third-party agencies for them to hire. Unemployed workers *including those who separated* search for work. Hiring agencies *observe whether a worker was endogenously separated* and choose whom to interview. The interviews reveal workers' productivity.

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- ⑤ Production occurs, and wages are paid.

Labor market: intermediate goods firms

- Intermediate firms of mass 1 operate in a competitive market and produce output using only labor.

$$X_t = A_t \underbrace{\psi_t}_{\text{avg. productivity}} \underbrace{N_t}_{\text{\# of employed workers}}$$

- Firms set **type-specific** hiring and firing thresholds $\bar{a}_{k,t}$, $\underline{a}_{k,t}$.
- Can rewrite output as

$$X_t = A_t \left\{ \frac{\gamma}{1 - \underline{s}} \left[(1 - \delta) \int_{\underline{a}_{h,t}}^1 a \, da + \delta \int_{\bar{a}_{h,t}}^1 a \, da \right] + \frac{1 - \gamma}{\bar{s}} \left[(1 - \delta) \int_{\underline{a}_{l,t}}^1 a \, da + \delta \int_{\bar{a}_{l,t}}^{\bar{s}} a \, da \right] \right\}$$

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I beleive this is missing.

Labor market: threshold determination

- Firms have full bargaining power.
- At the firing threshold $\underline{a}_{k,t}$, the firm is indifferent between firing and not firing the marginal worker so we have

$$W_{k,t} = \underbrace{P_t^I A_t \underline{a}_{k,t}}_{\text{MRPL}} + \underbrace{V_{k,t}}_{\text{option value}},$$

where P_t^I is the price of intermediate goods and $V_{k,t}$ is the option value of retaining the worker. formulas

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- Similarly, at the hiring thresholds $\bar{a}_{k,t}$, we have

$$\frac{G_l}{1 - \frac{\bar{a}_{l,t}}{\bar{s}}} + W_{l,t} = P_t^I A_t \bar{a}_{l,t} + V_{l,t}$$
$$\underbrace{\frac{G_h}{1 - \frac{\bar{a}_{h,t} - \underline{s}}{1 - \underline{s}}}}_{\text{expected hiring cost}} + W_{h,t} = P_t^I A_t \bar{a}_{h,t} + V_{h,t}$$

Labor market: hiring

- The firm specifies hiring and firing thresholds ($\bar{a}_{k,t}$, $\underline{a}_{k,t}$), and the agency chooses workers to interview accordingly.
 - Since $\bar{a}_{k,t} > \underline{a}_{k,t}$ in equilibrium, endogenously separated workers will not be interviewed.

- The monetary cost of interviewing a worker is assumed to be

$$G_{k,t} = F \cdot W_{k,t}$$

- Competitive agency markets $\Rightarrow$ firms pay the expected cost $G_{k,t}/P(a_{i,t} > \bar{a}_{k,t})$ for each worker hired, and every worker with productivity above the hiring threshold is hired.
- Hiring costs are removed from the economy.

Employment dynamics

- Unemployment level immediately after exogenous separation:

$$U_{h,t} = \gamma - (1 - \delta)N_{h,t-1}$$

$$U_{l,t} = 1 - \gamma - (1 - \delta)N_{l,t-1}$$

Employment dynamics

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- Total employment:

$$N_{h,t} = \left(1 - \frac{a_{h,t} - \underline{s}}{1 - \underline{s}}\right) (1 - \delta)N_{h,t-1} + \left(1 - \frac{a_{h,t} - \underline{s}}{1 - \underline{s}}\right) U_{h,t}$$

$$N_{l,t} = \underbrace{\left(1 - \frac{a_{l,t}}{\bar{s}}\right)}_{=P(a_{i,t} > \bar{a}_{k,t})} (1 - \delta)N_{l,t-1} + \underbrace{\left(1 - \frac{a_{l,t}}{\bar{s}}\right)}_{\text{mass of newly hired workers}} U_{l,t}$$

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- Simplified:

$$\begin{aligned}N_{h,t} &= \frac{1}{1 - \underline{s}} [\gamma(1 - \bar{a}_{h,t}) + (1 - \delta)(\bar{a}_{h,t} - \underline{a}_{h,t})N_{h,t-1}] \\N_{l,t} &= \frac{1}{\bar{s}} [(1 - \gamma)(\bar{s} - \bar{a}_{l,t}) + (1 - \delta)(\bar{a}_{l,t} - \underline{a}_{l,t})N_{l,t-1}]\end{aligned}$$

Rest of the model

- **Final goods firm.** CES monopolistic competition + Calvo pricing + linear technology $Y_t(i) = X_t(i)$. NKPC:

$$\pi_t = \lambda(p_t^i - p_t) + \beta \mathbb{E}_t[\pi_{t+1}]$$

- **Monetary policy.** Interest-rate smoothing as in Coibion and Gorodnichenko (2012):

$$i_t^* = \phi_\pi \pi_t + \phi_y y_t + \mu_t$$

$$\dot{i}_t = (1 - \rho_i) i_t^* + \rho_i \dot{i}_{t-1}$$

$$\mu_t = \rho_\mu \mu_{t-1} + \epsilon_t$$

Table 6: Calibration Parameters

Notes. This table reports the baseline parameters values.

$\beta = 0.99$	quarterly discount factor
$\sigma = 1$	inverse Intertemporal Elasticity of Substitution
$\chi = 4$	Disutility of working
$\mathbb{h} = 0.8$	Habit formation
$\theta = 0.73$	Calvo parameter
$\phi_{\pi} = 1.24$	Taylor rule response to interest rate
$\phi_y = 0.33/4$	Taylor rule response to output
$\rho_i = 0.7$	Interest rate smoothing
$\rho_{\mu} = 0.1$	Interest rate shock persistence
$F = 0.25$	Hiring cost
$\delta = 0.05$	Exogenous separation rate
$\bar{a}_h = 0.45$	Steady state hiring threshold H
$\bar{a}_l = 0.45$	Steady state hiring threshold L
$\underline{s} = 0.1$	Lower bound on support of productivity of high type
$\bar{s} = 0.75$	Upper bound on support of productivity of low type
$\gamma = 0.5$	Share of high types

Simulation exercise

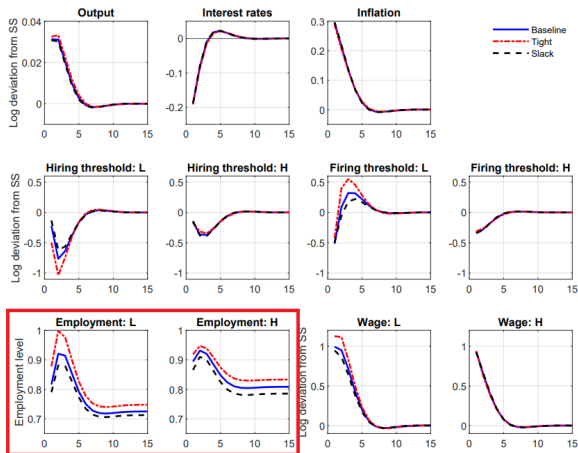
Goal. See how the IRFs of employment vary with labor market tightness.

- ① Shock = 1-SD decrease in interest rate
- ② Compute the IRFs for three different steady-states with different degrees of labor market tightness (i.e. employment).
- ③ Different steady-state levels of employment are achieved by varying the supports of labor productivity:

	$\underline{s}$	$\bar{s}$	high-type employment	low-type employment
Baseline	0.1	0.75	80.9%	72.5%
Tight	0.125	0.725	83.3%	74.8%
Slack	0.075	0.775	78.5%	71.2%

Simulation results

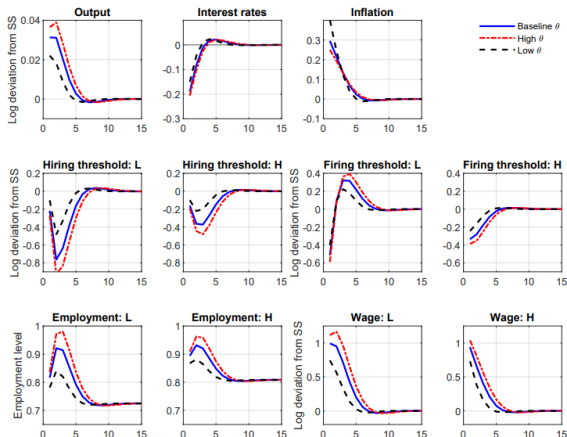
Figure 4: Impulse Response Functions for Different Steady-State Employment Levels



This figure plots impulse response functions for output, interest rate, inflation, hiring thresholds, firing thresholds, the fractions of workers employed, and wages for high (H) and low (L) type workers. Response functions are plotted for three different levels of the steady-state employment, that is, labor market tightness.

Simulation results: flatter NKPC

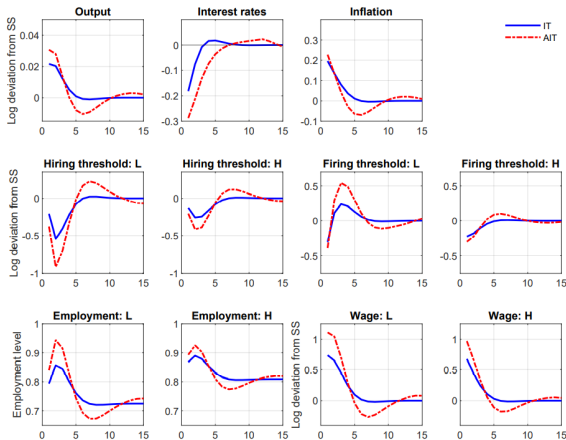
Figure 6: Impulse Response Functions for Different Degrees of Price Stickiness



This figure plots impulse response functions for output, interest rate, inflation, hiring thresholds, firing thresholds, the fractions of workers employed, and wages for high (H) and low (L) type workers. Response functions are plotted for different levels of price stickiness: the level of price stickiness, θ , is 0.73 in the baseline simulation (blue line), $e^{-1/4}$ in the high stickiness simulation (red line), and $e^{-1/2}$ in the low stickiness simulation.

Simulation results: average inflation targeting

Figure 7: Impulse Response Functions for Taylor Rule versus Average Inflation Targeting



This figure plots impulse response functions for output, interest rate, inflation, hiring thresholds, firing thresholds, the fractions of workers employed, and wages for high (H) and low (L) type workers. Response functions are plotted for a standard Taylor rule with interest rate smoothing and for a version with average inflation targeting adding eleven lags of inflation.

Conclusion

- Monetary expansions have stronger effect on the employment growth of less-attached workers (i.e. Blacks, high school non-completers, females) in tight labor markets.
- Such incremental effect is not found for more-attached workers.
- Need tighter markets to promote employment growth of less-attachment workers $\Rightarrow$ inflation-"broad-based employment" tradeoff

Some comments

Sample selection concerns me:

- Oversampling of MSAs with *more* less-attached workers and industrys with *more* less-attached workers.
- Want to see the geographical distribution of labor market tightness – where is the source of variation ?
- Robustness test: use the same sample for cross-group comparison.

Some comments

Empirical exercises exploit cross-sectional variation but do not account for other regional differences:

$$\begin{aligned} EmplGrowth_{j,g,m,t} = & \beta_1 \times FedFunds_t \times Empl/Pop_{m,t-1} \\ & + \gamma \times FedFunds_t \times Confounder_{m,t-1} + \dots \end{aligned}$$

One example is housing equity (Beraja, Fuster, Hurst, Vavra 2019).

- Neither industry-location FE nor industry-time FE fix this.
- Quick (semi-)fix: include location-time interactive FE (i.e. exploit only time-series variation in the tightness)
- Better: include controls interacted with federal funds rate.

Some comments

I am unsure how to interpret the main result, which matters for policy recommendation.

- ① MP less effective in MSA with low employment-to-population ratio
 - These MSA have low natural employment. Why bother to stimulate?
- ② MP less effective in periods when the employment-to-population ratio is low
 - Should include location-time interactive FE then.

Construction of high-frequency monetary shocks

- 1 Calculate the surprise component of the announced change in the federal funds rate as

$$v_t = \frac{D}{D - t} (ff_{t+\Delta t^+,0} - ff_{t-\Delta t^-,0}),$$

where t is the time when the FOMC issues an announcement, $ff_{t+\Delta t^+,0}$ is the fed funds futures rate 20 minutes after t , $ff_{t-\Delta t^-,0}$ is the fed funds futures rate 10 minutes before t , and D is the number of days in the month.

- 2 When the event day occurs within the last seven days of the month, follow Gürkaynak et al. (2005) and use the unscaled change in the next-month futures contract.
- 3 For each quarter t , sum the innovations that occurred from the start of the sample period through t .

Labor market: extra formulas

- The price of intermediate goods P_t^I is given by

$$P_t^I = \gamma W_{h,t} \left[1 - \frac{\underline{a}_{h,t} - \underline{s}}{1 - \underline{s}} - \delta \frac{\bar{a}_{h,t} - \underline{a}_{h,t}}{1 - \underline{s}} \right] \\ + (1 - \gamma) W_{l,t} \left[1 - \frac{\underline{a}_{l,t}}{\bar{s}} - \delta \frac{\bar{a}_{l,t} - \underline{a}_{l,t}}{\bar{s}} \right]$$

- The option value of retaining a worker $V_{k,t}$ is given by

$$V_{h,t} = \beta(1 - \delta) \mathbb{E}_t \left[\frac{Z_{t+1}}{Z_t} \left[\left(1 - \frac{\underline{a}_{h,t+1} - \underline{s}}{1 - \underline{s}} \right) (G_{h,t+1} + V_{h,t+1}) \right] \right] \\ V_{l,t} = \beta(1 - \delta) \mathbb{E}_t \left[\frac{Z_{t+1}}{Z_t} \left[\left(1 - \frac{\underline{a}_{l,t+1}}{\bar{s}} \right) (G_{l,t+1} + V_{l,t+1}) \right] \right]$$

- $V_{k,t+1}$ is set to 0 in simulation.